

Equivalence Relations

Section 9.5

Section Summary

- Equivalence Relations (等价关系)
- Equivalence Classes (等价类)
- Equivalence Classes and Partitions

Equivalence Relations

Definition 1: A relation on a set A is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

Definition 2: Two elements a , and b that are related by an equivalence relation are called *equivalent*. The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

Strings

Example: Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Is R an equivalence relation?

Solution: Show that all of the properties of an equivalence relation hold.

- *Reflexivity:* Because $l(a) = l(a)$, it follows that aRa for all strings a .
- *Symmetry:* Suppose that aRb . Since $l(a) = l(b)$, $l(b) = l(a)$ also holds and bRa .
- *Transitivity:* Suppose that aRb and bRc . Since $l(a) = l(b)$, and $l(b) = l(c)$, $l(a) = l(c)$ also holds and aRc .

Congruence Modulo m

Example: Let m be an integer with $m > 1$. Show that the relation
 $R = \{(a,b) \mid a \equiv b \pmod{m}\}$
 is an equivalence relation on the set of integers.

Solution: Recall that $a \equiv b \pmod{m}$ if and only if m divides $a - b$.

- *Reflexivity:* $a \equiv a \pmod{m}$ since $a - a = 0$ is divisible by m since $0 = 0 \cdot m$.
- *Symmetry:* Suppose that $a \equiv b \pmod{m}$. Then $a - b$ is divisible by m , and so $a - b = km$, where k is an integer. It follows that $b - a = (-k)m$, so $b \equiv a \pmod{m}$.
- *Transitivity:* Suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. Then m divides both $a - b$ and $b - c$. Hence, there are integers k and l with $a - b = km$ and $b - c = lm$. We obtain by adding the equations:

$$a - c = (a - b) + (b - c) = km + lm = (k + l)m.$$
 Therefore, $a \equiv c \pmod{m}$.

Divides

Example: Show that the “divides” relation on the set of positive integers is not an equivalence relation.

Solution: The properties of reflexivity, and transitivity do hold, but there relation is not transitive. Hence, “divides” is not an equivalence relation.

- *Reflexivity:* $a \mid a$ for all a .
- *Not Symmetric:* For example, $2 \mid 4$, but $4 \nmid 2$. Hence, the relation is not symmetric.
- *Transitivity:* Suppose that a divides b and b divides c . Then there are positive integers k and l such that $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c . Therefore, the relation is transitive.

Equivalence Classes

Definition 3: Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the *equivalence class* of a . The equivalence class of a with respect to R is denoted by $[a]_R$.

When only one relation is under consideration, we can write $[a]$, without the subscript R , for this equivalence class.

Note that $[a]_R = \{s \mid (a,s) \in R\}$.

- If $b \in [a]_R$, then b is called a representative (代表元) of this equivalence class. Any element of a class can be used as a representative of the class.
- The equivalence classes of the relation congruence modulo m are called the *congruence classes modulo m* . The congruence class of an integer a modulo m is denoted by $[a]_m$, so $[a]_m = \{\dots, a-2m, a-m, a, a+m, a+2m, \dots\}$. For example,

$$[0]_4 = \{\dots, -8, -4, 0, 4, 8, \dots\}$$

$$[1]_4 = \{\dots, -7, -3, 1, 5, 9, \dots\}$$

$$[2]_4 = \{\dots, -6, -2, 2, 6, 10, \dots\}$$

$$[3]_4 = \{\dots, -5, -1, 3, 7, 11, \dots\}$$

Equivalence Classes and Partitions

Theorem 1: let R be an equivalence relation on a set A . These statements for elements a and b of A are equivalent:

- (i) aRb
- (ii) $[a] = [b]$
- (iii) $[a] \cap [b] \neq \emptyset$

Proof: We show that (i) implies (ii). Assume that aRb . Now suppose that $c \in [a]$. Then aRc . Because aRb and R is symmetric, bRa . Because R is transitive and bRa and aRc , it follows that bRc . Hence, $c \in [b]$. Therefore, $[a] \subseteq [b]$. A similar argument (omitted here) shows that $[b] \subseteq [a]$. Since $[a] \subseteq [b]$ and $[b] \subseteq [a]$, we have shown that $[a] = [b]$.

■ Show that (2) implies (3)

$$[a] = [b]$$

R is reflexive $\Rightarrow [a]$ is nonempty

$$\left. \begin{array}{l} [a] = [b] \\ R \text{ is reflexive} \Rightarrow [a] \text{ is nonempty} \end{array} \right\} \Rightarrow [a] \cap [b] \neq \emptyset$$

$$(1) aRb$$

$$(2) [a] = [b]$$

$$(3) [a] \cap [b] \neq \emptyset$$

■ Show that (3) implies (1)

$$[a] \cap [b] \neq \emptyset \Rightarrow \exists x \in [a] \cap [b]$$

$$\Rightarrow (a, x) \in R, (b, x) \in R$$

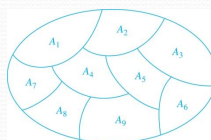
$$\Rightarrow (a, x) \in R, (x, b) \in R$$

$$\Rightarrow (a, b) \in R$$

Partition of a Set

Definition: A *partition* of a set S is a collection of disjoint nonempty subsets of S that have S as their union. In other words, the collection of subsets A_i , where $i \in I$ (where I is an index set), forms a partition of S if and only if

- $A_i \neq \emptyset$ for $i \in I$,
- $A_i \cap A_j = \emptyset$ when $i \neq j$,
- and $\bigcup_{i \in I} A_i = S$.



Notation:

$$pr(A) = \{A_i \mid i \in I\}$$

A Partition of a Set

An Equivalence Relation Partitions a Set

- Let R be an equivalence relation on a set A . The union of all the equivalence classes of R is all of A , since an element a of A is in its own equivalence class $[a]_R$. In other words,

$$\bigcup_{a \in A} [a]_R = A.$$

- From Theorem 1, it follows that these equivalence classes are either equal or disjoint, so $[a]_R \cap [b]_R = \emptyset$ when $[a]_R \neq [b]_R$.
- Therefore, the equivalence classes form a partition of A , because they split A into disjoint subsets.

An Equivalence Relation Partitions a Set (*continued*)

Theorem 2: Let R be an equivalence relation on a set S . Then the equivalence classes of R form a partition of S . Conversely, given a partition $\{A_i \mid i \in I\}$ of the set S , there is an equivalence relation R that has the sets A_i , $i \in I$, as its equivalence classes.

Proof: We have already shown the first part of the theorem.

For the second part, assume that $\{A_i \mid i \in I\}$ is a partition of S . Let R be the relation on S consisting of the pairs (x, y) where x and y belong to the same subset A_i in the partition. We must show that R satisfies the properties of an equivalence relation.

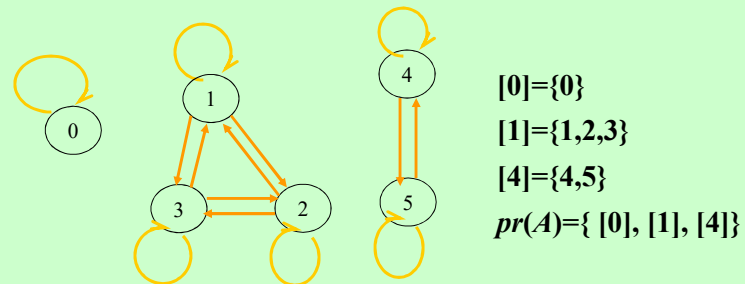
- Reflexivity:** For every $a \in S$, $(a, a) \in R$, because a is in the same subset as itself.
- Symmetry:** If $(a, b) \in R$, then b and a are in the same subset of the partition, so $(b, a) \in R$.
- Transitivity:** If $(a, b) \in R$ and $(b, c) \in R$, then a and b are in the same subset of the partition, as are b and c . Since the subsets are disjoint and b belongs to both, the two subsets of the partition must be identical. Therefore, $(a, c) \in R$ since a and c belong to the same subset of the partition.

[[Example]] Find the partition of the set A from R .

$$A = \{0, 1, 2, 3, 4, 5\},$$

$$R = \{(0,0), (1,1), (2,2), (3,3), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (4,4), (4,5), (5,4), (5,5)\}$$

Solution:



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Question:

Congruence Modulo m

$$R = \{(a, b) \mid a \equiv b \pmod{m}, a, b \in \mathbb{Z}\}, pr(\mathbb{Z}) = ?$$

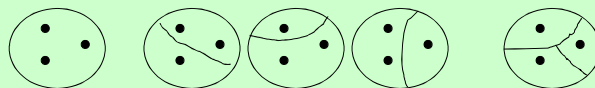
$$pr(\mathbb{Z}) = \{[0]_m, [1]_m, \dots, [m-1]_m\}$$

Question:

$|A|=3$. How many different equivalence relations on the set A are there?

Solution:

an equivalence relation on a set $A \leftrightarrow$ a partition of A



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The combining of the Equivalence Relations

- Let R and S be equivalence relations on A , how about $R \cap S$, $R \cup S$?
- The answer for $R \cap S$ is yes.
- (1) Is it reflexive?:
- Since $(a,a) \in R$ $(a,a) \in S$, then $(a,a) \in R \cap S$ for every $a \in A$
- (2) Is it symmetric?
- Since $R^{-1}=R$ $S^{-1}=S$, then $(R \cap S)^{-1} = R^{-1} \cap S^{-1} = R \cap S$

The combining of the Equivalence Relations

- Is it transitive?
- Since R and S are transitive, then $R^2 \subseteq R$, $S^2 \subseteq S$
- Then $(R \cap S)^2 = R^2 \cap R \circ S \cap S \circ R \cap S^2 \subseteq R^2 \cap S^2 = R \cap S$
- So $R \cap S$ is equivalence relation.
- But the answer for $R \cup S$ is No!

【 Theorem 】 If R_1, R_2 are equivalence relations on A , then $R_1 \cup R_2$ is a reflexive and symmetric relation on A .

Proof:

(1) reflexive

$$\forall a \in A \quad \because (a, a) \in R_1, (a, a) \in R_2 \quad \therefore (a, a) \in R_1 \cup R_2$$

(2) symmetric

$$(a, b) \in R_1 \cup R_2 \Rightarrow (a, b) \in R_1 \text{ or } (a, b) \in R_2$$

$$\Rightarrow (b, a) \in R_1 \text{ or } (b, a) \in R_2 \Rightarrow (b, a) \in R_1 \cup R_2$$

Question: transitive?

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9.5 Equivalence Relations

【 Example 】 $A = \{a, b, c\}$,

$$R_1 = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$$

$$R_2 = \{(a, a), (b, b), (c, c), (b, c), (c, b)\}$$

Is $R_1 \cup R_2$ a transitive relation ?

Solution:

$$R_1 \cup R_2 = \{(a, a), (b, b), (c, c), (a, b), (b, a), (b, c), (c, b)\}$$

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9.5 Equivalence Relations

【 Theorem 】 If R_1, R_2 are equivalence relations on A , then $(R_1 \cup R_2)^*$ is an equivalence relation on A .

Proof:

- (1) reflexive**
- (2) symmetric**
- (3) transitive**

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Homework

- **Sec. 9.5** 3, 10, 16, 36(b), 39, 41

Partial Orderings

Section 9.6

Section Summary

- Partial Orderings (偏序) and Partially-ordered Sets
- Lexicographic Orderings
- Hasse Diagrams
- Lattices (格)
- Topological Sorting

Partial Orderings

Definition 1: A relation R on a set S is called a *partial ordering*, or *partial order*, if it is reflexive, antisymmetric, and transitive. A set together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R) . Members of S are called *elements* of the poset.

Partial Orderings (*continued*)

Example 1: Show that the “greater than or equal” relation ($\geq$) is a partial ordering on the set of integers.

- *Reflexivity:* $a \geq a$ for every integer a .
- *Antisymmetry:* If $a \geq b$ and $b \geq a$, then $a = b$.
- *Transitivity:* If $a \geq b$ and $b \geq c$, then $a \geq c$.

These properties all follow from the order axioms for the integers.
(See Appendix 1).

Partial Orderings (*continued*)

Example 2: Show that the divisibility relation ($|$) is a partial ordering on the set of integers.

- *Reflexivity:* $a | a$ for all integers a . (see Example 9 in Section 9.1)
- *Antisymmetry:* If a and b are positive integers with $a | b$ and $b | a$, then $a = b$. (see Example 12 in Section 9.1)
- *Transitivity:* Suppose that a divides b and b divides c . Then there are positive integers k and l such that $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c . Therefore, the relation is transitive.
- $(\mathbb{Z}^+, |)$ is a poset.

Partial Orderings (*continued*)

Example 3: Show that the inclusion relation ($\subseteq$) is a partial ordering on the power set of a set S .

- *Reflexivity:* $A \subseteq A$ whenever A is a subset of S .
- *Antisymmetry:* If A and B are positive integers with $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- *Transitivity:* If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

The properties all follow from the definition of set inclusion.

Comparability

Definition 2: The elements a and b of a poset $(S, \leq)$ are *comparable* if either $a \leq b$ or $b \leq a$. When a and b are elements of S so that neither $a \leq b$ nor $b \leq a$, then a and b are called *incomparable*.

The symbol $\leq$ is used to denote the relation in any poset.

Definition 3: If $(S, \leq)$ is a poset and every two elements of S are comparable, S is called a *totally ordered* or *linearly ordered set*, and $\leq$ is called a *total order* (全序) or a *linear order* (线序). A totally ordered set is also called a *chain* (链).

Definition 4: $(S, \leq)$ is a *well-ordered* (良序) set if it is a poset such that $\leq$ is a total ordering and every nonempty subset of S has a least element.

Lexicographic Order

Definition: Given two posets $(A_1, \leq_1)$ and $(A_2, \leq_2)$, the *lexicographic ordering* on $A_1 \times A_2$ is defined by specifying that (a_1, a_2) is less than (b_1, b_2) , that is,

$$(a_1, a_2) < (b_1, b_2),$$

either if $a_1 <_1 b_1$ or if $a_1 = b_1$ and $a_2 <_2 b_2$.

- This definition can be easily extended to a lexicographic ordering on strings (see text).

Example: Consider strings of lowercase English letters. A lexicographic ordering can be defined using the ordering of the letters in the alphabet. This is the same ordering as that used in dictionaries.

- $discreet < discrete$, because these strings differ in the seventh position and $e < t$.
- $discreet < discreteness$, because the first eight letters agree, but the second string is longer.

[[Example] Let $A_1 = A_2 = \mathbb{Z}^+$ and $R_1 = R_2 = |$. Then

(1) $(2, 4) < (2, 8)$ since $x_1 = x_2$ and $y_1 R_2 y_2$

(2) $(2, 4)$ is not related under R to $(2, 6)$ since $x_1 = x_2$ but 4 does not divide 6.

(3) $(2, 4) < (4, 5)$ since $x_1 R_1 x_2$

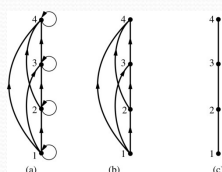
$(2, 3, 4, 5) < (2, 3, 8, 2)$ since $x_1 = x_2$ and $y_1 = y_2$ and 4 divides 8.

$(2, 3, 4, 5)$ is not related to $(3, 6, 8, 10)$ since 2 does not divide 3.

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Hasse Diagrams

Definition: A *Hasse diagram* is a visual representation of a partial ordering that leaves out edges that must be present because of the reflexive and transitive properties.

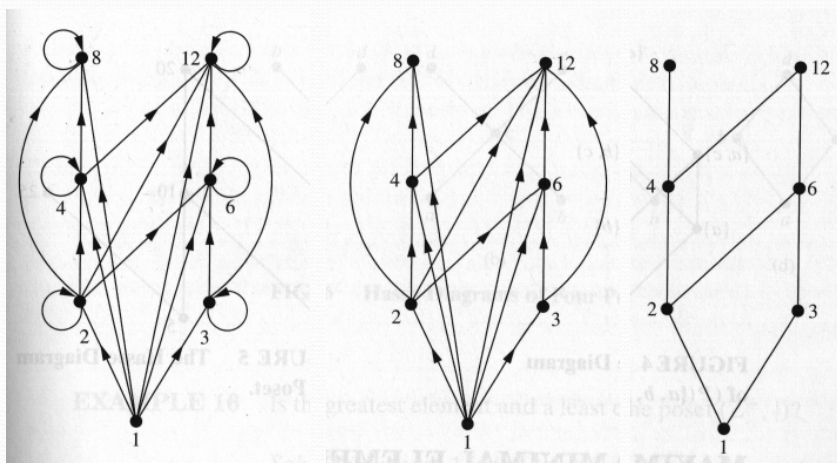


A partial ordering is shown in (a) of the figure above. The loops due to the reflexive property are deleted in (b). The edges that must be present due to the transitive property are deleted in (c). The Hasse diagram for the partial ordering (a), is depicted in (c).

Procedure for Constructing a Hasse Diagram

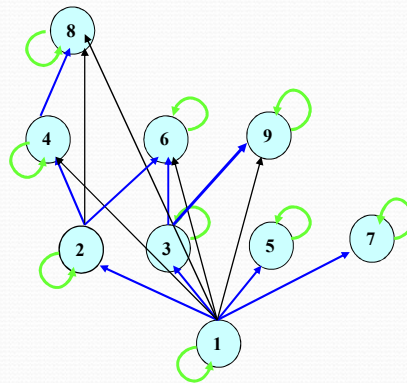
- To represent a finite poset $(S, \leq)$ using a Hasse diagram, start with the directed graph of the relation:
 - Remove the loops (a, a) present at every vertex due to the reflexive property.
 - Remove all edges (x, y) for which there is an element $z \in S$ such that $x < z$ and $z < y$. These are the edges that must be present due to the transitive property.
 - Arrange each edge so that its initial vertex is below the terminal vertex. Remove all the arrows, because all edges point upwards toward their terminal vertex.

For the poset $(\{1, 2, 3, 4, 6, 8, 12\}, |)$



9.6 Partial Orderings

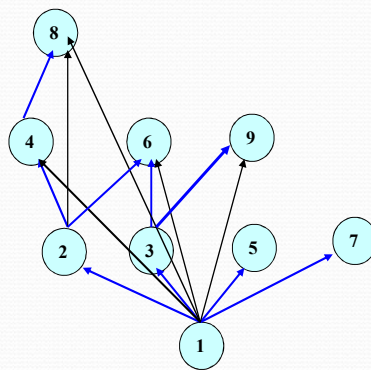
[[Example]] $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ $R = \{(a, b) \mid a \mid b, a, b \in A\}$



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9.6 Partial Orderings

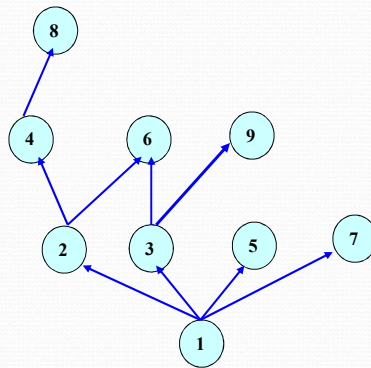
[[Example]] $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ $R = \{(a, b) \mid a \mid b, a, b \in A\}$



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9.6 Partial Orderings

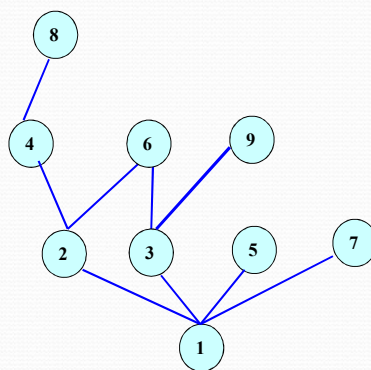
[[Example]] $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ $R = \{(a, b) \mid a \mid b, a, b \in A\}$



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9.6 Partial Orderings

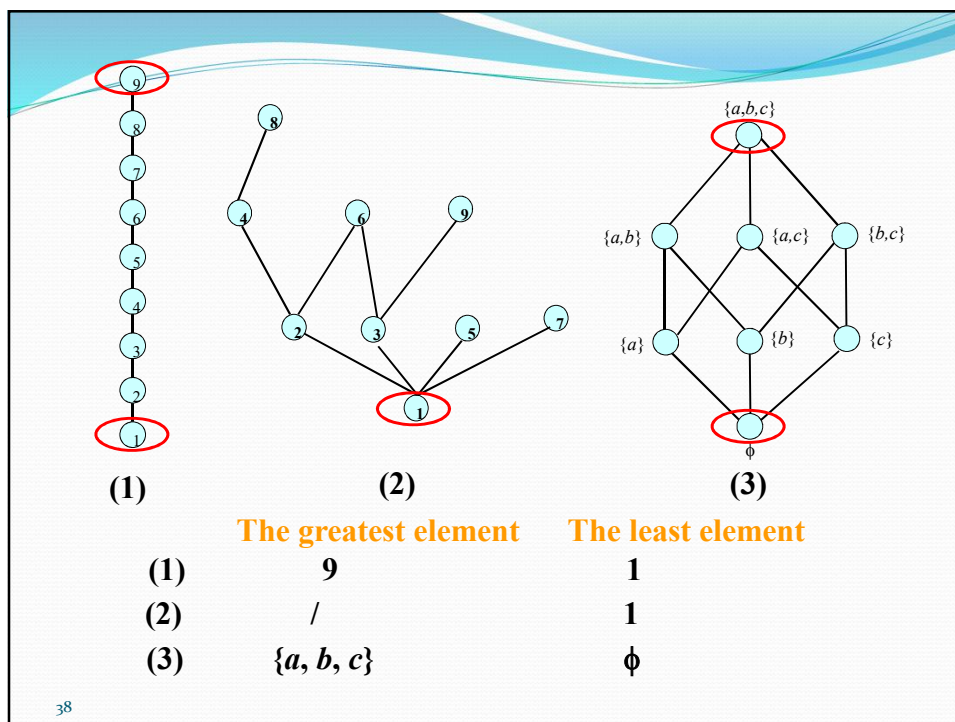
[[Example]] $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ $R = \{(a, b) \mid a \mid b, a, b \in A\}$



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Hasse Diagram Terminology

- Let $(S, \preceq)$ be a poset.
- a is *maximal* (极大元) in $(S, \preceq)$ if there is no $b \in S$ such that $a \preceq b$. (**top** of the Hasse diagram)
- a is *minimal* (极小元) in $(S, \preceq)$ if there is no $b \in S$ such that $b \preceq a$. (**bottom** of the Hasse diagram)
- a is the *greatest element* (最大元) of $(S, \preceq)$ if $b \preceq a$ for all $b \in S$.
- a is the *least element* (最小元) of $(S, \preceq)$ if $a \preceq b$ for all $b \in S$.



【 Theorem 】 The greatest and least element of the poset $(A, \leq)$ are unique when they exist.

Proof:

Suppose that a_1 is a greatest element in A . It follows that

$x \leq a_1$ for every x in A .

Suppose that a_2 is a greatest element in A . It follows that

$x \leq a_2$ for every x in A .

It implies that $a_2 \leq a_1$ and $a_1 \leq a_2$

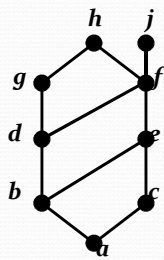
That is $a_1 = a_2$

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Hasse Diagram Terminology (Cont ..)

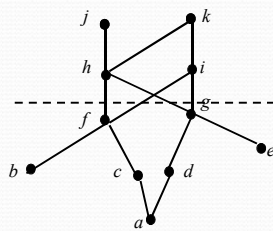
- Let A be a subset of $(S, \preceq)$.
- If $u \in S$ such that $a \preceq u$ for all $a \in A$, then u is called an *upper bound* of A .
- If $l \in S$ such that $l \preceq a$ for all $a \in A$, then l is called an *lower bound* of A .
- If x is an upper bound of A and $x \preceq z$ whenever z is an upper bound of A , then x is called the *least upper bound* of A .
- If y is a lower bound of A and $z \preceq y$ whenever z is a lower bound of A , then y is called the *greatest lower bound* of A .

Example



Maximal elements: h, j
 Minimal elements: a
 Greatest element: none
 Least element: a
 Upper bound of $\{a, b, c\}$: $\{e, f, j, h\}$
 Least upper bound of $\{a, b, c\}$: e
 Lower bound of $\{a, b, c\}$: $\{a\}$
 Greatest lower bound of $\{a, b, c\}$: a

【Example】 $S = \{a, b, c, d, e, f, g, h, i, j, k\}$
 $A = \{a, b, c, d, e, f, g\}, A' = \{h, i, j, k\}$



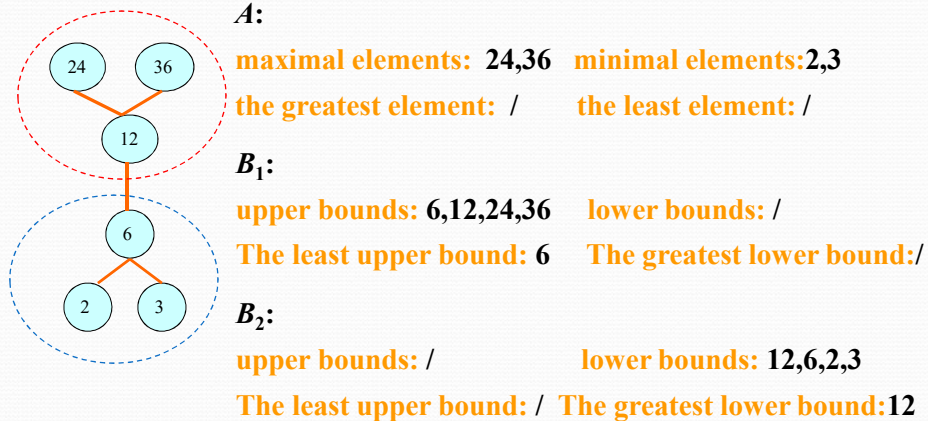
the upper bounds of set A : h, i, k, j

the lower bounds of set A : /

the upper bounds of set A' : /

the lower bounds of set A' : f, g, a, b, c, d, e

【Example】 $A = \{2, 3, 6, 12, 24, 36\}$, $B_1 = \{2, 3, 6\}$, $B_2 = \{12, 24, 36\}$, $R : |$
Determine the maximal elements, minimal elements, greatest element, least element of set A , the upper bounds, lower bounds, least upper bound, greatest lower bound of set B_1, B_2 .



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Lattices

- A partially ordered set in which every pair of elements has both a least upper bound and greatest lower bound is called a *lattice*.

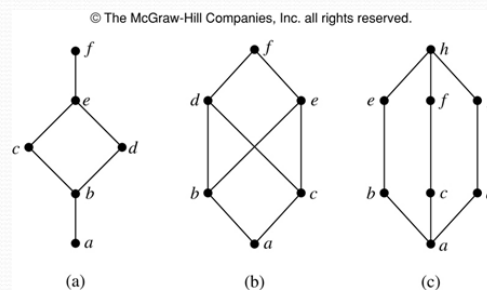


FIGURE 8 Hasse Diagrams of Three Posets.

9.6 Partial Orderings

(2) $(\mathbb{Z}, \leq)$?

lub: the larger of the two elements,

glb: the smaller of the two elements.

Hence, the poset $(\mathbb{Z}, \leq)$ is a lattice.

Note: Every totally ordered set is a lattice.

(3) $(\mathbb{Z}^+, |)$?

$\forall a, b \in \mathbb{Z}^+$

lub: the least common multiple,

glb: the greatest common divisor

Hence, the poset $(\mathbb{Z}^+, |)$ is a lattice.

(4) $(P(s), \subseteq)$?

$\forall s_1, s_2 \in P(s)$, **lub:** $s_1 \cup s_2$ **glb:** $s_1 \cap s_2$

Hence, the poset $(P(s), \subseteq)$ is a lattice.

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Topological Sorting

- A total ordering $\preceq$ is said to be compatible with the partial ordering R if $a \preceq b$ whenever $a R b$.
 - Topological sorting: constructing a compatible total ordering from a partial ordering
- Lemma 1: Every finite nonempty poset $(S, \preceq)$ has at least one minimal element.

Topological Sorting

- Algorithm 1: topological sorting
 - Procedure topological sort($S, \preceq$: finite poset)
 - $k := 1$
 - while $S \neq \emptyset$
 - begin
 - $a_k :=$ a minimal element of S
 - $S := S - \{a_k\}$
 - $k := k + 1$
 - end
 - (See Fig. 9, Fig. 10, 11)

FIGURE 9 (9.6,p629)

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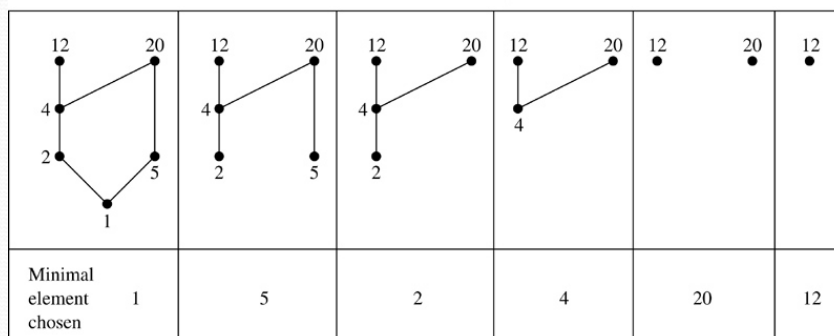


FIGURE 9 A Topological Sort of $(\{1, 2, 4, 5, 12, 20\}, \preceq)$.

P. 577

FIGURE 10 (9.6)

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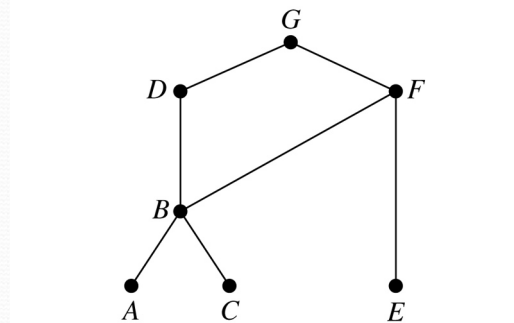


FIGURE 10 The Hasse Diagram for Seven Tasks.

FIGURE 11 (9.6)

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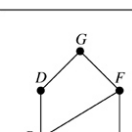
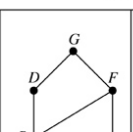
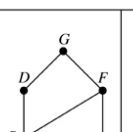
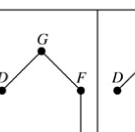
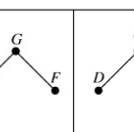
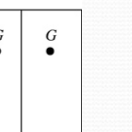
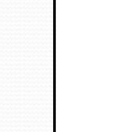
							
Minimal element chosen	A	C	B	E	F	D	G

FIGURE 11 A Topological Sort of the Tasks.

Homework

Sec. 9.6 5, 10, 23(a),(c), 32, 44, 66